Take-Home Exercise: Predicting Football Match Corners

Scenario

You are given a synthetic dataset of football match statistics spanning seasons 2012–2022. Each row represents one team's performance in a single match. The target variable is `corners` – the number of corner kicks taken by that team.

Build the best model you can for predicting `corners` in future matches – that is, matches played after the data you have been given.

Dataset

File: `match_data.csv` (~4,000 rows, ~40 teams, 11 seasons)

Column	Description
`match_id`	Unique match identifier (two rows per match: one per team)
`date`	Match date
`season`	Season year (2012–2022)
`team`	Team identifier
`opponent`	Opponent identifier
`home`	1 if playing at home, 0 if away
`corners`	Number of corners (integer; this is the target variable)
`feature_1` ... `feature_10`	Explanatory variables (continuous)

Assume the feature values for a given game would have been available at the time of prediction, i.e you can use the value of all the features in row i to model corners in row i.

Holdout predictions

File: `match_data_holdout_features.csv` – a held-back set of matches from seasons *after* the training data. Same schema as `match_data.csv` minus the `corners` column.

Your code must produce predictions for every row in this file and write them to `predictions.csv` with columns:

Column	Description
`match_id`	Copied from the holdout file
`team`	Copied from the holdout file (each match has two rows, one per team)
`predicted_corners`	Your model's prediction

Note: `match_id` alone is **not** unique — every match has two rows, one per team. You must include both `match_id` and `team` so each prediction can be matched to the correct row.

Baseline

`baseline_validation_predictions.csv` contains reference-model predictions for a recent labeled validation slice of `match_data.csv`, keyed by `match_id` and `team`, with a `baseline_predicted_corners` column. Use it to make a comparison to the reference baseline. Treat the file as comparison-only: do not use it as a model input or copy it into `predictions.csv`.

Deliverables

Please submit:

- **Code** with clear instructions for reproducing your results (we will run it)
- **A written report** containing:
  - Your final model written in mathematical notation
  - A **decision log** covering key choices, including dead ends and changes of direction
  - **What next**: what you would investigate if given another week
- **AI usage appendix** (if applicable): you may use any tools including AI assistants. If you do, include a short appendix describing what you delegated, what you modified or rejected, and why. Absence of this appendix is fine if you did not use AI.

Evaluation

We will assess the quality of your analysis, code, and written communication. There is no single correct answer; we care about your reasoning process and how you support your conclusions.